Forensics Analytics

Graph based Cryptography Fraud detection using ML

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Abstract

Cryptocurrency transactions are recorded on public blockchains, making transaction information available for analysis; however, the pseudonymous nature and complex flow of digital assets can make the investigation of fraudulent and illicit transactions difficult.

This project proposes an Explainable Graph-Based Cryptocurrency Fraud Investigation Framework for detecting suspicious Bitcoin transactions and analyzing their associated money-flow patterns. The proposed system uses the publicly available Ethereum Dataset, where transactions are represented as nodes and transaction relationships are represented as directed edges.

The transaction network is first preprocessed and enriched with transaction-level and graph-based features such as transaction frequency, connectivity, degree and centrality measures.

Abstract

Traditional machine learning models such as Random Forest and XGBoost are used as baseline classifiers and compared with a Graph Convolutional Network for capturing relationships between connected transactions.

After detecting potentially illicit transactions, the system performs suspicious transaction path analysis to trace the movement of funds through connected transactions and identify potentially significant nodes and subgraphs.

Explainable AI techniques such as SHAP and GNNExplainer are incorporated to identify the features, neighboring transactions and graph structures that influence the model's prediction.

The final system provides a forensic risk score, suspicious transaction ranking, transaction-flow vis. The proposed approach aims to support investigators by combining fraud detection, graph-based transaction analysis and explainable artificial intelligence into a single cryptocurrency forensic analytics framework.



Approach

Data Engineering

- Ethereum Dataset: 10,000+ addresses, 50+ features
- Graph Feature Engineering: Created 5 behavioral metrics:
  - Volume Ratio, ERC-20 Ratio, Unique Address Ratio
  - Contract Patterns, Value Distribution Analysis

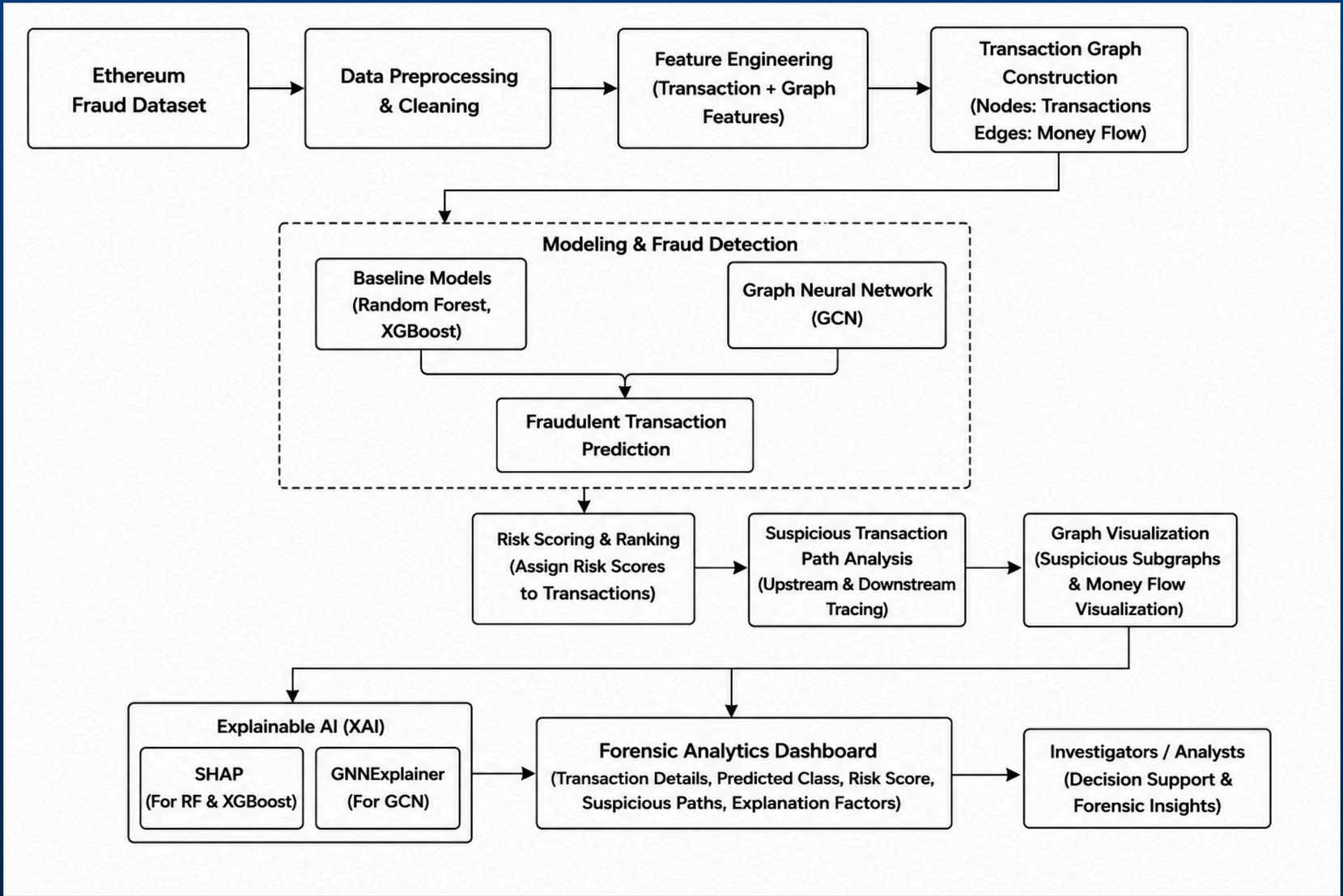
Model Development

- Algorithm: Random Forest Classifier
- Handling: Upsampling for class imbalance
- Performance: ROC-AUC > 0.85, High Precision/Recall

Novelty - Explainable Investigation

- SHAP Analysis: Identifies WHY addresses are flagged
- Path Tracing: Visualizes suspicious money flows
- Interpretable Output: Provides actionable intelligence for investigators

Block Diagram



TITLE	METHODS	DATASET	MAIN IDEA
A Novel Graph-Based Framework for Cryptocurrency Fraud Detection	GNN, Graph Fusion, Graph Reasoning	Cryptocurrency transaction data	Uses transaction graphs and GNNs to detect fraudulent entities and transactions.
Cryptocurrency Fraud Detection through Classification Techniques	Logistic Regression, RF, KNN, AdaBoost, XGBoost	Ethereum transaction dataset	Compares ML classifiers for detecting fraudulent Ethereum transactions.
Enhancing Digital Finance Security: AI-Based Approaches for Credit Card and Cryptocurrency Fraud Detection	XGBoost, LightGBM, CNN, Autoencoder, SMOTE	Financial & cryptocurrency transactions	Applies AI models and SMOTE for effective fraud detection under class imbalance.

TITLE	METHODS	DATASET	MAIN IDEA
Deep Learning for Fraud Detection in Bitcoin Transactions: An Artificial Intelligence-Based Strategy	ANN, RNN, CNN, Autoencoder	Existing/secondary data	Explores deep learning for detecting abnormal and fraudulent Bitcoin activity.
Forensic Analysis of Cryptocurrency Transactions: Leveraging Blockchain for Fraud Detection and Regulatory Compliance	Blockchain analytics, AI monitoring	Blockchain transaction data / literature	Focuses on blockchain-based forensic investigation, fraud detection and compliance.
Secure and Scalable Blockchain-Based Federated Learning for Cryptocurrency Fraud Detection: A Systematic Review	Federated Learning, Blockchain, Differential Privacy	Existing research	Reviews privacy-preserving and scalable approaches to crypto fraud detection.